

Project Design Phase-II

Data Flow Diagram & User Stories

Date	25 September 2026
Team ID	_____
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, how information is processed, and where data is stored. The following Level-0 DFD represents the proposed Automated Network Request Management system in ServiceNow.

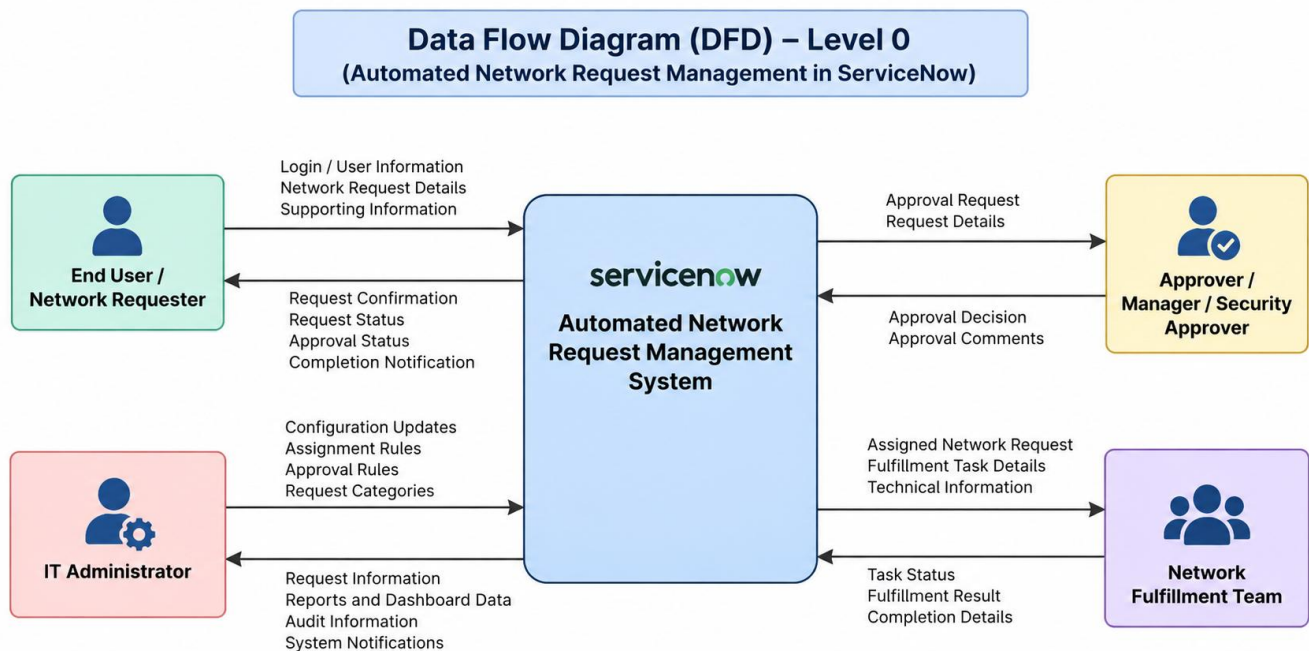


Figure 1: Data Flow Diagram (DFD) – Level 0

Level-0 Data Flow Description

End User / Network Requester: Submits user information, network request details and supporting information; receives confirmation, status, approval and completion notifications.

Approver: Receives approval requests and request details; sends approval decisions and comments.

Network Fulfillment Team: Receives assigned requests and task details; updates task status and fulfillment results.

IT Administrator: Maintains request categories, assignment and approval rules; receives reports, dashboard and audit information.

User Stories

The following user stories translate the functional requirements of the proposed system into actionable user-focused requirements.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
End User / Requester	Network Request Submission	USN-1	As a requester, I can access the Network Request Catalog so that I can select the required network service.	Request categories are visible and a request type can be selected.	High	Sprint-1
End User / Requester	Network Request Submission	USN-2	As a requester, I can submit a network request by entering the required request information.	Mandatory information can be entered and the request submitted.	High	Sprint-1
End User / Requester	Request Form & Validation	USN-3	As a requester, I can see mandatory fields and validation messages so that I can provide complete request information.	Required fields are identified and incomplete requests cannot be submitted.	High	Sprint-1
End User / Requester	Request Tracking & Notifications	USN-4	As a requester, I can view the current status of my network request.	Relevant lifecycle status is visible.	High	Sprint-1
End User / Requester	Request Tracking & Notifications	USN-5	As a requester, I can receive notifications when my request status changes.	Relevant submission, approval, fulfillment and closure notifications are generated.	Medium	Sprint-2
Approver	Approval Management	USN-6	As an approver, I can view network requests that require my approval.	Pending approval requests are visible to the authorized approver.	High	Sprint-1
Approver	Approval Management	USN-7	As an approver, I can approve or reject a network request.	The approver can approve or reject and provide comments when required.	High	Sprint-1
Approver	Approval Management	USN-8	As an approver, I can receive notifications for requests requiring my approval.	The approver receives relevant approval notifications.	Medium	Sprint-2
Network Fulfillment Team	Automated Assignment & Fulfillment	USN-9	As a network team member, I can view network requests assigned to my fulfillment group.	Assigned requests are available to authorized team members.	High	Sprint-1
Network Fulfillment Team	Automated Assignment & Fulfillment	USN-10	As a network team member, I can view the technical details required to fulfill a network request.	Relevant request information is available from the task.	High	Sprint-1
Network Fulfillment Team	Automated Assignment & Fulfillment	USN-11	As a network team member, I can update the status of a fulfillment task.	Task status and details can be updated.	High	Sprint-2

Network Fulfillment Team	Automated Assignment & Fulfillment	USN-12	As a network team member, I can record the completion details of a network request.	Completion information is saved and the request can proceed to closure.	Medium	Sprint-2
IT Administrator	Request Management & Administration	USN-13	As an IT administrator, I can manage network request categories and catalog items.	Authorized administrators can configure request types.	High	Sprint-1
IT Administrator	Request Management & Administration	USN-14	As an IT administrator, I can configure approval and assignment rules.	Authorized administrators can define approvers and fulfillment groups.	High	Sprint-2
IT Administrator	Audit & Reporting	USN-15	As an IT administrator, I can monitor network requests using reports or dashboards.	The administrator can view request status and fulfillment activity.	Medium	Sprint-2
IT Administrator	Audit & Reporting	USN-16	As an IT administrator, I can view request history and audit information.	Authorized administrators can review request, approval and fulfillment history.	Medium	Sprint-2

User Story to Functional Requirement Mapping

Functional Requirement	Related User Stories
FR-1 Network Request Submission	USN-1, USN-2
FR-2 Request Form & Validation	USN-3
FR-3 Approval Management	USN-6, USN-7, USN-8
FR-4 Automated Assignment & Fulfillment	USN-9, USN-10, USN-11, USN-12
FR-5 Request Tracking & Notifications	USN-4, USN-5
FR-6 Request Management & Administration	USN-13, USN-14
FR-7 Audit & Reporting	USN-15, USN-16